

EXPECTED LEARNING OUTCOMES:

The purpose of this exercise is to compare diagnostic features in several types of tumors in CNS and to correlate the clinical and morphologic features.

After this CASES laboratory the student will be able to do the following:

1. Describe the relative frequency of primary and metastatic CNS tumors in adult population.
2. Recognize that the most common neoplasm in the CNS is not of CNS origin.
3. List the neoplasms that most commonly spread into the CNS.
4. Recognize the characteristic presentation of metastatic neoplasms to the brain
5. Describe the general clinical and morphologic features of benign primary CNS tumors and their potential complications:
 - Acoustic neuroma (vestibular Schwannoma)
 - Meningioma
 - Choroid plexus papilloma
6. Describe the general clinical and morphologic features of malignant primary neoplasms in the CNS in adults and their potential complications
 - Anaplastic astrocytoma (WHO grade III)
 - Glioblastoma multiforme (WHO grade IV)
 - Primary CNS lymphoma

Recommended pre-lab tasks:

1. Go through the corresponding lecture PP file.
2. Go through the CNS tumors REVIEW CASES PP files. There is a PP file of the review cases both without and with a KEY.
3. Read Robbins and Kumar Basic Pathology 11th Ed.: Central Nervous System and Eye: Tumors.

PROCEDURE:

Time	Activity
70 minutes	Study and discuss the review cases with your group. Check the key for the correct answers. You do not need to submit anything on these cases.
20 minutes	Q & A on the review cases
60 minutes	Finish studying the three unknown cases with your group and submit a learning response by 11.30 am using the LR form. You do not need to submit a response to the questions on the slides.

	The team members may submit the same learning response for the unknown cases, but each team member must submit it individually to get the points for this exercise.
20 minutes	Discussion of the unknown cases. We can start this earlier, as soon as everybody has submitted their learning response.

KEY WORDS/ CONCEPTS

Acoustic neuroma Anaplastic astrocytoma Antoni A area Antoni B area Anaplastic astrocytoma Astrocytoma Choroid plexus papilloma CNS lymphoma, primary vs. secondary Ependymoma HIF-1α Glioblastoma multiforme (GBM) Glomeruloid bodies	Medulloblastoma Meningioma Metastatic adenocarcinoma Neurofibromatosis type 1 and 2 Primary CNS lymphoma Primary vs. secondary GBM Pseudo-palisading Schwannoma Supratentorial vs. infratentorial Verocay bodies Vascular endothelial growth factor (VEGF)
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REVIEW CASES:

SPECIMEN	TISSUE	CLINICAL DIAGNOSIS	PATHOLOGIC DIAGNOSIS
Case 1 1345.svs	Intradural, extra-axial mass in the cerebello-pontine angle	Acoustic neuroma	Acoustic neuroma (Vestibular Schwannoma), benign
Case 2 758.svs	Left parietal region Intracranial mass	Brain tumor	Meningioma, syncytial type, benign
Case 3 1344.svs	Brain mass, floor of the 4 th ventricle	Brain tumor	Choroid plexus papilloma, benign
Case 4 1346.svs	Left frontotemporal brain mass	Malignant brain tumor, S/P treatment for Astrocytoma WHO Grade II	Anaplastic astrocytoma (WHO Grade III)
Case 5 1343.svs	Right supratentorial brain mass	Malignant brain tumor, S/ P resection and radio- and chemotherapy	Glioblastoma multiforme (Astrocytoma WHO grade IV)
Case 6 753.svs	Right supratentorial brain mass	Brain tumor	Primary B-cell lymphoma

There are two PP files, one labeled 2023-10-19-ISMSN-CNSTUMORS-REVCASES and a second one labeled KEY-2023-10-19-ISMSN-CNSTUMORS-REVCASES, which contains discussion of the histologic features.

The Unknown Cases PP file and the Learning Response form will be available on 10.19.23 at 9.30 am in the Assignments folder in Canvas.

Please submit your learning response by 10.30 am on 10-19-23 using the LR form that is provided in the Assignments folder.

STUDY QUESTIONS (to be answered on your own time):

- 1. What is the most common neoplasm in the CNS?**
- 2. What are the common features of primary versus metastatic neoplasms in the CNS?**
- 3. Which genetic disease is associated with multiple fibroblastic meningiomas?**
- 4. Why do meningiomas grow rapidly during pregnancy?**
- 5. Name 3 primary malignant CNS tumors arising from 3 different cell types.**
- 6. Which microscopic features indicate advancing grade of malignancy in astrocytomas?**
- 7. What is the difference between astrocytoma grade III and glioblastoma multiforme (astrocytoma grade IV)?**
- 8. What are the characteristic features of pilocytic astrocytomas as to patient population, tumor location, microscopic features, and clinical behavior?**
- 9. What causes the appearance of “pseudopalisading” in glioblastomas?**
- 10. What are glomeruloid bodies?**
- 11. List 3 characteristic features of ependymomas.**
- 12. List 3 characteristic features of oligodendrogliomas.**
- 13. Name a highly malignant posterior fossa tumor in young children.**
- 14. What are Verocay bodies?**